CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

# Chapter 5: Arithmetic Progressions

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Common difference and the nth term

**1.** Find the common difference of the AP $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \ldots$ (a) $\sqrt{2}$ (b) 1 (c) $2\sqrt{2}$ (d) $-\sqrt{2}$ CBSE 2026, 1 mark

SOLUTION

$d = a_2 - a_1 = 2\sqrt{2} - \sqrt{2} = \sqrt{2}$

**Answer: (a)**

**2.** The numbers $p - 1$, $p + 1$ and $2p + 3$ are in AP. Find $p$. (a) $-2$ (b) 4 (c) 0 (d) 2 CBSE 2023, 1 mark

SOLUTION

In an AP the middle term is the average of its neighbours: $2(p + 1) = (p - 1) + (2p + 3)$.

$2p + 2 = 3p + 2$, so $p = 0$.

**Answer: (c)**

**3.** How many two-digit numbers are divisible by 3? CBSE 2019, 1 mark

SOLUTION

They are $12, 15, 18, \ldots, 99$: an AP with $a = 12$, $d = 3$ and last term 99.

$99 = 12 + (n - 1) \times 3$, so $n - 1 = 29$

**Answer: 30**

**4.** Which term of the AP $3, 15, 27, 39, \ldots$ is 120 more than its 21st term? CBSE 2019, 2 marks

SOLUTION

$a = 3$, $d = 12$. $a_{21} = 3 + 20 \times 12 = 243$, so the required term is $243 + 120 = 363$.

$3 + (n - 1) \times 12 = 363$, so $n - 1 = 30$

**Answer: the 31st term**

(Shortcut: 120 more means $120 \div 12 = 10$ terms further on, so it is term $21 + 10 = 31$.)

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**5.** In an AP the common difference is $-4$ and the seventh term is 4. Find the first term. CBSE 2018, 1 mark

SOLUTION

$a_7 = a + 6d$: $4 = a + 6(-4)$, so $a = 4 + 24$.

**Answer:** $a = 28$

**6.** Four consecutive terms of an AP add up to 32. The product of the first and last of them is to the product of the two middle ones as $7 : 15$. Find the four numbers. CBSE 2018, 4 marks

SOLUTION

Let the numbers be $a - 3d$, $a - d$, $a + d$, $a + 3d$. Their sum is $4a = 32$, so $a = 8$.

$$\frac{(a-3d)(a+3d)}{(a-d)(a+d)} = \frac{64 - 9d^2}{64 - d^2} = \frac{7}{15}$$

$960 - 135d^2 = 448 - 7d^2$, so $128d^2 = 512$, $d^2 = 4$ and $d = \pm 2$.

With $d = 2$ the numbers are 2, 6, 10, 14. With $d = -2$ they are the same numbers in reverse order.

**Answer: 2, 6, 10, 14**

**7.** For what value of $k$ are $k + 9$, $2k - 1$ and $2k + 7$ consecutive terms of an AP? CBSE 2016, 1 mark

SOLUTION

The middle term is the average of its neighbours: $2(2k - 1) = (k + 9) + (2k + 7)$

$4k - 2 = 3k + 16$

**Answer:** $k = 18$

**8.** The 4th term of an AP is zero. Prove that its 25th term is three times its 11th term. CBSE 2016, 2 marks

SOLUTION

$a_4 = a + 3d = 0$, so $a = -3d$.

$a_{25} = a + 24d = 21d$ and $a_{11} = a + 10d = 7d$

**So $a_{25} = 3 \times a_{11}$. Proved.**

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## 2 Case study

**9.** In a potato race, 10 potatoes lie in a straight line. The first is 5 m from a bucket at the starting point and the rest are 3 m apart. A runner starts at the bucket, fetches the nearest potato, runs back to drop it in the bucket, and repeats this for each potato in turn. (i) How far does the runner go to collect the first potato? (ii) How far to collect the second? (iii) Find the total distance run, OR the time taken for the whole race at an average speed of 5 m/s. CBSE 2026, 4 marks

SOLUTION

Each trip is to the potato and back, so it is twice the distance of that potato from the bucket.

(i) $2 \times 5$. **First potato: 10 m.**

(ii) The second potato is $5 + 3 = 8$ m away: $2 \times 8$. **Second potato: 16 m.**

(iii) The trips are $10, 16, 22, \ldots$, an AP with $a = 10$, $d = 6$ and $n = 10$.

$S_{10} = \frac{10}{2}[2(10) + 9(6)] = 5 \times 74$. **Total distance: 370 m.**

OR: time $= \frac{\text{distance}}{\text{speed}} = \frac{370}{5}$. **Time taken: 74 seconds.**

**10.** In a fund-raising run, the first round is 300 m long and every round after it is 50 m longer than the one before, so the rounds are 300 m, 350 m, 400 m and so on. There are 10 rounds in all. (i) Write the 4th, 5th and 6th terms of this AP. (ii) How long is the 8th round? (iii) Find the total distance covered in all 10 rounds, OR the total distance covered by a runner who stops after 6 rounds. CBSE 2025, 4 marks

SOLUTION

The lengths form an AP with $a = 300$ and $d = 50$.

(i) $a_4 = 300 + 3(50) = 450$, $a_5 = 500$, $a_6 = 550$. **The terms are 450, 500 and 550.**

(ii) $a_8 = 300 + 7(50) = 650$. **The 8th round is 650 m.**

(iii) $S_{10} = \frac{10}{2}[2(300) + 9(50)] = 5 \times 1050 = 5250$. **Total for 10 rounds: 5250 m.**

OR: $S_6 = \frac{6}{2}[2(300) + 5(50)] = 3 \times 850 = 2550$. **Total for 6 rounds: 2550 m.**

**11.** A matchstick pattern is built in stages. Figure 1 is a square with a triangle on each of its four sides. Each new figure adds one more square to the row, with triangles on its free sides. Figures 1, 2 and 3 contain 4, 6 and 8 triangles, and use 12, 19 and 26 matchsticks. (i) Write the AP for the number of triangles and its $n$th term. (ii) Which figure uses 61 matchsticks? CBSE 2022 Term 2, 4 marks

SOLUTION

(i) The numbers of triangles are $4, 6, 8, \ldots$ with $a = 4$ and $d = 2$. $a_n = 4 + (n-1) \times 2$. $a_n = 2n + 2$

(ii) The numbers of matchsticks are $12, 19, 26, \ldots$ with $a = 12$ and $d = 7$.

$12 + (n-1) \times 7 = 61$, so $7(n-1) = 49$ and $n = 8$. **Figure 8 uses 61 matchsticks.**

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## 3 Sum of the first n terms

**12.** The sum of the first $m$ terms of an AP is $2m^2 + 3m$. Find its second term. CBSE 2025, 1 mark

SOLUTION

$S_1 = 2 + 3 = 5$, so the first term is 5.

$S_2 = 2(4) + 3(2) = 14$

Second term $= S_2 - S_1 = 14 - 5$

**Answer: 9**

**13.** An A.P. has first term $a = 7$ and $n$th term $a_n = 84$, and its first $n$ terms add up to $\frac{2093}{2}$. Find $n$. (a) 22 (b) 24 (c) 23 (d) 26 CBSE 2024, 1 mark

SOLUTION

$S_n = \frac{n}{2}(a + a_n) = \frac{n}{2}(7 + 84) = \frac{91n}{2}$

$\frac{91n}{2} = \frac{2093}{2}$ gives $n = 23$

**Answer: (c)**

**14.** In an A.P., the first and eighth terms have sum 32 and product 60. Find the first term and the common difference, and then the sum of the first 20 terms. CBSE 2024, 5 marks

SOLUTION

$a + (a + 7d) = 32$, so $a + 7d = 32 - a$.

Product: $a(32 - a) = 60$, so $a^2 - 32a + 60 = 0$ and $(a - 2)(a - 30) = 0$. Hence $a = 2$ or $a = 30$.

If $a = 2$: $7d = 28$, so $d = 4$. Then $S_{20} = \frac{20}{2}[\,2(2) + 19(4)\,] = 10 \times 80 = 800$.

If $a = 30$: $7d = -28$, so $d = -4$. Then $S_{20} = \frac{20}{2}[\,2(30) + 19(-4)\,] = 10 \times (-16) = -160$.

**Answer:** $a = 2$, $d = 4$, $S_{20} = 800$; **or** $a = 30$, $d = -4$, $S_{20} = -160$.

**15.** An A.P. has 40 terms. Its first 9 terms add up to 153 and its last 6 terms add up to 687. Find the first term, the common difference and the sum of all 40 terms. CBSE 2024, 5 marks

SOLUTION

$S_9 = \frac{9}{2}[\,2a + 8d\,] = 153$, so $a + 4d = 17$.

The last 6 terms are $a_{35}$ to $a_{40}$. Their sum $= \frac{6}{2}(a_{35} + a_{40}) = 3(2a + 73d) = 687$, so $2a + 73d = 229$.

Substituting $a = 17 - 4d$: $34 - 8d + 73d = 229$, so $65d = 195$ and $d = 3$. Then $a = 5$.

$S_{40} = \frac{40}{2}[\,2(5) + 39(3)\,] = 20 \times 127 = 2540$

**Answer:** $a = 5$, $d = 3$, **and the sum of all 40 terms is 2540.**

**16.** In an AP, the 11th term and the 17th term are in the ratio $3:4$. Find the ratio of the 5th term to the 21st term, and the ratio of the sum of the first 5 terms to the sum of the first 21 terms. CBSE 2023, 5 marks

SOLUTION

$\frac{a+10d}{a+16d}=\frac{3}{4}$, so $4a+40d=3a+48d$ and $a=8d$.

$\frac{a_5}{a_{21}}=\frac{a+4d}{a+20d}=\frac{12d}{28d}=\frac{3}{7}$

$S_5=\frac{5}{2}(2a+4d)=\frac{5}{2}(20d)=50d$ and $S_{21}=\frac{21}{2}(2a+20d)=\frac{21}{2}(36d)=378d$

$\frac{S_5}{S_{21}}=\frac{50}{378}=\frac{25}{189}$

**Answer:** $a_5:a_{21}=3:7$ **and** $S_5:S_{21}=25:189$

**17.** 250 logs are stacked in rows: 22 in the bottom row, 21 in the next, 20 in the one above, and so on. How many rows are there, and how many logs are in the top row? CBSE 2023, 5 marks

SOLUTION

The rows form an AP with $a=22$ and $d=-1$, and $S_n=250$.

$\frac{n}{2}[44-(n-1)]=250$, so $n(45-n)=500$

$n^2-45n+500=0$, so $(n-20)(n-25)=0$ and $n=20$ or $25$.

If $n=25$, the top row would have $22-24=-2$ logs, which is impossible. So $n=20$.

Top row $=a_{20}=22-19=3$

**Answer: 20 rows, with 3 logs in the top row**

**18.** Find the sum of the first 30 terms of the AP $-30,-24,-18,\ldots$ CBSE 2022 Term 2, 2 marks

SOLUTION

$a=-30, d=6, n=30$

$S_{30}=\frac{30}{2}[2(-30)+29(6)]=15\times 114$

**Answer: 1710**

**19.** The sum of the first $n$ terms of an AP is $S_n=n(4n+1)$. Find the AP. CBSE 2022 Term 2, 2 marks

SOLUTION

$S_1=1\times 5=5$, so $a_1=5$.

$S_2=2\times 9=18$, so $a_2=S_2-S_1=13$ and $d=13-5=8$.

**Answer:** $5,13,21,29,\ldots$

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**20.** The sum of the first $n$ terms of an AP is $S_n = 3n^2 - 4n$. Find its $n$th term. CBSE 2019, 2 marks

SOLUTION

$a_n = S_n - S_{n-1} = (3n^2 - 4n) - [3(n-1)^2 - 4(n-1)]$

$= 3n^2 - 4n - (3n^2 - 10n + 7)$

**Answer:** $a_n = 6n - 7$

**21.** The first four terms of an AP add up to 40 and the first 14 terms add up to 280. Find the sum of its first $n$ terms. CBSE 2019, 4 marks

SOLUTION

$S_4 = 2(2a + 3d) = 40$, so $2a + 3d = 20$ ... (1)

$S_{14} = 7(2a + 13d) = 280$, so $2a + 13d = 40$ ... (2)

Subtracting (1) from (2): $10d = 20$, so $d = 2$ and $a = 7$.

$S_n = \frac{n}{2}[14 + 2(n-1)] = \frac{n}{2}(2n + 12)$

**Answer:** $S_n = n^2 + 6n$

**22.** Find the sum of the first 8 multiples of 3. CBSE 2018, 2 marks

SOLUTION

The multiples are $3, 6, 9, \ldots, 24$: an AP with $a = 3$, $l = 24$ and $n = 8$.

$S_8 = \frac{8}{2}(3 + 24) = 4 \times 27$

**Answer: 108**

**23.** The sums of the first $n$ terms of two APs are in the ratio $(7n + 1) : (4n + 27)$. Find the ratio of their $m$th terms. CBSE 2016, 3 marks

SOLUTION

Let the APs have first terms $a$, $A$ and common differences $d$, $D$. Then $\frac{2a + (n-1)d}{2A + (n-1)D} = \frac{7n + 1}{4n + 27}$.

The ratio of the $m$th terms is $\frac{a + (m-1)d}{A + (m-1)D} = \frac{2a + (2m-2)d}{2A + (2m-2)D}$, which is the left side above with $n - 1 = 2m - 2$, that is $n = 2m - 1$.

Put $n = 2m - 1$: $\frac{7(2m-1) + 1}{4(2m-1) + 27}$

**Answer:** $(14m - 6) : (8m + 23)$

**24.** The houses in a row are numbered from 1 to 49. Show that there is a house number $X$ for which the sum of the numbers before it equals the sum of the numbers after it. CBSE 2016, 4 marks

SOLUTION

Sum of the numbers before $X$: $1 + 2 + \ldots + (X-1) = \dfrac{X(X-1)}{2}$

Sum of the numbers after $X$: $S_{49} - S_X = \dfrac{49 \times 50}{2} - \dfrac{X(X+1)}{2}$

Equating: $X(X-1) = 2450 - X(X+1)$, so $2X^2 = 2450$ and $X^2 = 1225$.

$X = 35$, which is a whole number between 1 and 49.

**Answer: such a house exists, and it is number 35**

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